Elijah LaMothe

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EDUCATION

University of California, Berkeley — B.S. Mechanical Engineering, Minor in EECS *Expected May 2029*

GPA: 3.5

Relevant Coursework: Physics 7B (Electricity & Magnetism), APE DeCal (FPGA Digital Oscilloscope Design), ENGIN 7 (Python & Numerical Methods), Math 53 (Multivariable Calculus), ME 40 (Thermodynamics), ENGIN 26 (SolidWorks CAD)

HARDWARE & RESEARCH EXPERIENCE

UC Berkeley RoboSub — Vice President; Electrical Lead *2025 – Present*

- Lead the electrical team in a ground-up redesign of the AUV's electrical architecture for the 2026–27 vehicle, replacing a monolithic power-distribution board with a modular stacked-card system for high-current thruster loads under a $1,000 budget.
- Defined the architecture and authored top-level hierarchical KiCad schematics for a fully passive power-distribution card and an adapter card; onboarded ~8 new members via an intro PCB design project and organized them into per-card design teams.
- Directing a 3-person sub-team designing a dedicated STM32-based safety board in Altium Designer that monitors onboard sensors and automatically cuts ESC power on fault conditions, providing a redundant shutdown path alongside the diver kill switch; set up a Git-based workflow so the team can version, review, and merge the shared Altium project.
- Replacing the vehicle's commercial Pixhawk flight controller with a custom ESP32/STM32-based controller and contributing to its firmware, including thruster voltage limits enforced in firmware to protect the power system and thrusters.
- Previously Software/Controls Co-Lead: developed attitude and depth control software and behavior-tree mission logic, and began a custom Unity-based simulator modeling vehicle dynamics; also designed the camera, electronics, and battery housings.

TAFLab (Theoretical and Applied Fluids Lab), UC Berkeley — Undergraduate Researcher *Summer 2026*

- Built and field-tested autonomous ocean-monitoring drones measuring water temperature, conductivity, and wave height to support predictive maps of hazardous wave conditions, enabling safer and more fuel-efficient ship routing.
- Improved the vessel's sail mechanism, increasing mounting consistency and structural stability through iterative field tests.

PROJECTS

FPGA-Based Digital Oscilloscope — APE DeCal, UC Berkeley *Fall 2026 – Present*

- Building a digital oscilloscope from scratch with a student team, using an FPGA to route sampled signal data in real time.
- Gaining hands-on experience with digital logic, signal acquisition, and bench-level hardware bring-up and debugging.

HOPCopter — Custom Autonomous Quadcopter *2026 – Present*

- Designed the electronics architecture and a custom power-distribution PCB in KiCad for a 4S battery-powered quadcopter, including regulated low-voltage power rails for the ESP32 flight computer, IMU, and time-of-flight sensors.
- Developing embedded firmware for sensor integration, telemetry, and brushless motor/ESC control, with hands-on debugging.
- Working toward closed-loop flight using IMU-based state estimation and PID control, targeting autonomous altitude and position hold; designed the frame and electronics mounts in CAD and iterated on them through 3D printing.

Autonomous Dead-Wheel Odometry Tracking System *2023 – 2025*

- Designed encoder-based odometry dead-wheel housings and implemented PID and nonlinear motion control for autonomous navigation; the system contributed to a world-record competition season and a top-1% global ranking.

TECHNICAL SKILLS

Hardware & EDA: KiCad, Altium Designer, PCB design, ESP32, STM32, FPGA (intro), IMU/ToF/encoder sensors, ESCs
Programming: Python, Arduino/C-style embedded programming, Java, Kotlin, Git
Robotics & Mechanical: ROS2, PX4, Gazebo, Unity, PID control; Onshape, SolidWorks, 3D printing, rapid prototyping
Certifications: Certified Onshape Associate; Certified Onshape Professional

COMMUNITY

- Gears in Motion — Co-Coordinator: organize STEM/robotics outreach across 5 public libraries, reaching 600+ participants.